Tab 1

นายปริญญ์นกร อยู่แท้กูล 673380277-1

Excellent. We will now **strictly use your new topology and configuration**:

- ✅ R1 G0/0 → Internet (DHCP, NAT outside)
- ✅ R1–R2 connected via **Serial S0/0/0**
- ✅ ISP domain: **100.10.10.0/30**
- ✅ LAN A: 192.168.10.0/24
- ✅ LAN B: 192.168.20.0/24
- ✅ Microservices from **Week03 (Phase 1 Mockup)** FastAPI architecture

We now reframe this as:

🧪 LAB 5 — Internet Edge + ISP Serial WAN + Week03 Microservices

🎯 LAB OBJECTIVE

Transform Week03 microservices into a **realistic enterprise deployment**:

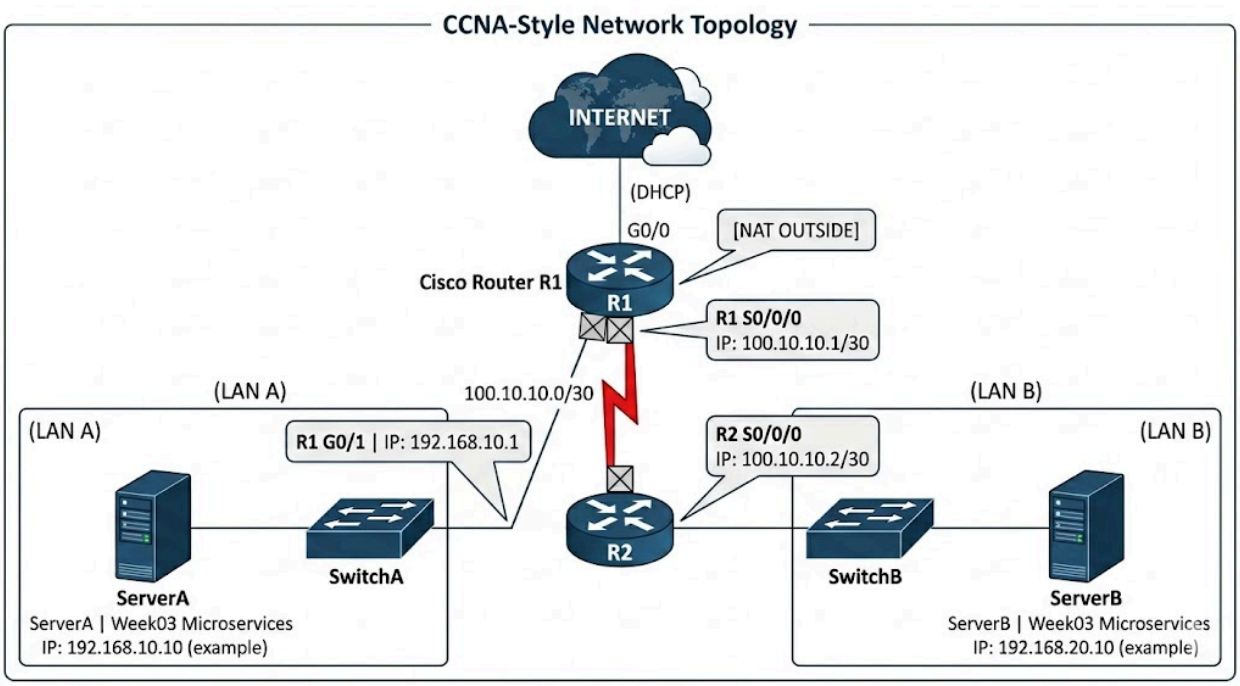
- R1 = Internet Edge Router (NAT + DHCP WAN)
- Serial WAN = ISP Domain (100.10.10.0/30)
- R2 = Branch Router
- Microservices deployed on ServerA and ServerB
- Validate cross-site microservice access over routed WAN + NAT

🗺️ LAB 5 TOPOLOGY (Reframed)

None

```
INTERNET
    |
(DHCP)
R1 G0/0
```

```
                    |
              [NAT OUTSIDE]
                    |
             R1 S0/0/0
          100.10.10.1/30
                    |
             R2 S0/0/0
          100.10.10.2/30
                    |
        -----------------------
        |                      |
R1 G0/1 (LAN A)       R2 G0/1 (LAN B)
192.168.10.1           192.168.20.1
        |                      |
    ServerA                ServerB
Week03 Microservices   Week03 Microservices
```



📍 ADDRESS TABLE (FINAL)

Device	Interface	IP	Role
R1	G0/0	DHCP	Internet
R1	S0/0/0	100.10.10.1/30	ISP
R1	G0/1	192.168.10.1/24	LAN A
R2	S0/0/0	100.10.10.2/30	ISP
R2	G0/1	192.168.20.1/24	LAN B
ServerA	eth0	192.168.10.10	Microservices A
ServerB	eth0	192.168.20.10	Microservices B

🔧 PART 1 — CISCO CONFIGURATION (STRICT)

🧱 R1 — Internet Edge Router

```
None
enable
conf t
hostname R1

interface g0/0
 ip address dhcp
 ip nat outside
 no shutdown

interface g0/1
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
```

```
 no shutdown

interface s0/0/0
 ip address 100.10.10.1 255.255.255.252
 clock rate 64000
 no shutdown

ip route 192.168.20.0 255.255.255.0 100.10.10.2
ip route 0.0.0.0 0.0.0.0 g0/0

access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload

end
write memory
```

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
R1(config)#interface g0/0
R1(config-if)# ip address dhcp
R1(config-if)# ip nat outside
R1(config-if)# no shutdown

R1(config-if)#
R1(config-if)#interface g0/1
R1(config-if)# ip address 192.168.10.1 255.255.255.0
R1(config-if)# ip nat inside
R1(config-if)# no shutdown

R1(config-if)#
R1(config-if)#interface s0/3/0
R1(config-if)# ip address 100.10.10.1 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
R1(config-if)#
R1(config-if)#exit
R1(config)#
R1(config)#ip route 192.168.20.0 255.255.255.0 100.10.10.2
R1(config)#ip route 0.0.0.0 0.0.0.0 g0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R1(config)#
R1(config)#access-list 10 permit 192.168.10.0 0.0.0.255
R1(config)#ip nat inside source list 10 interface g0/0 overload
R1(config)#
R1(config)#end
R1#write memory
Building configuration...
[OK]
R1#
```

Copy Paste

หน้าจอการตั้งค่า (CLI) ของ Router R1 ฝั่งสำนักงานใหญ่ ส่วนนี้ผมได้ทำการคอนฟิก IP Address, เซ็ตพอร์ต G0/0 ให้รับ IP แบบ DHCP เพื่อออกเน็ต, ตั้งค่า NAT (Inside/Outside) และชี้ Static Route เพื่อให้แพ็กเก็ตสามารถวิ่งข้ามไปยังวงแลนของฝั่งสาขาได้ถูกต้อง

🧱 R2 — Branch Router

```
None
enable
conf t
hostname R2

interface g0/1
 ip address 192.168.20.1 255.255.255.0
 no shutdown

interface s0/0/0
 ip address 100.10.10.2 255.255.255.252
 no shutdown

ip route 192.168.10.0 255.255.255.0 100.10.10.1
ip route 0.0.0.0 0.0.0.0 100.10.10.1

end
write memory
```

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no


Press RETURN to get started!



Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#
R2(config)#interface g0/1
R2(config-if)# ip address 192.168.20.1 255.255.255.0
R2(config-if)# no shutdown

R2(config-if)#
R2(config-if)#interface s0/3/0
R2(config-if)# ip address 100.10.10.2 255.255.255.252
R2(config-if)# no shutdown

R2(config-if)#
R2(config-if)#exit
R2(config)#
R2(config)#ip route 192.168.10.0 255.255.255.0 100.10.10.1
R2(config)#ip route 0.0.0.0 0.0.0.0 100.10.10.1
R2(config)#
R2(config)#end
R2#write memory
Building configuration...
[OK]
R2#
```

Copy Paste

สำหรับการตั้งค่า Router R2 ฝั่งสาขา ตัวนี้คอนฟิกจะไม่ซับซ้อนเท่า R1 ครับ ผมกำหนด IP ให้พอร์ตเชื่อมต่อปกติ และใช้การทำ Default Route (0.0.0.0 0.0.0.0) เพื่อโยนทราฟฟิกทั้งหมดที่ไม่ได้อยู่ในวงแลนตัวเอง กลับไปที่ R1 เสมอ ทำให้สาขาสามารถคุยกับเน็ตเวิร์กหลักและออกเน็ตได้

💻 PART 2 — Deploy Week03 Microservices

Based on your Week03 architecture from `PROJECT_PROFILE.md`

We deploy:

- Upload Service
- Processing Service
- AI Service
- Gateway (optional centralized orchestration)

🛠️ On ServerA (LAN A)

```shell
cd automation/week03-microservices/phase1
python start_services.py
```

Services:

- Upload → 8000
- Processing → 8001
- AI → 8002
- Gateway → 9000

```
PS C:\Users\computer\automation\week03-microservices\phase1> python start_services.py
🎯 Starting Phase 1 Microservices...
================================================
🚀 Starting Upload Service on port 8001...
⏳ Waiting for Upload Service to become ready...
❌ Upload Service process died
❌ Failed to start all services

🧹 Cleaning up...
✅ Stopped Upload Service
PS C:\Users\computer\automation\week03-microservices\phase1>
```

พอร์ต 8001 ที่ระบบต้องการใช้งาน อาจถูกโปรแกรมอื่นหรือโปรเซสเก่าที่ปิดไม่สนิทจองเอาไว้ ทำให้รันเซอร์วิสใหม่ไม่ได้

ServerA IP:

```
None
192.168.10.10
```

🛠️ On ServerB (LAN B)

Same deployment:

```
Shell
python start_services.py
```

ServerB IP:

```
None
192.168.20.10
```

🧪 PART 3 — WAN + INTERNET TESTING

🔵 Test 1 — Cross-Site Upload (LAN A → LAN B)

From ClientA:

```
None
curl http://192.168.20.10:8000/health
```

Should traverse:

ClientA → R1 → Serial ISP → R2 → ServerB

🔵 Test 2 — Full Workflow Across ISP

From LAN A client:

```
None
curl -X POST http://192.168.20.10:9000/process-file
```

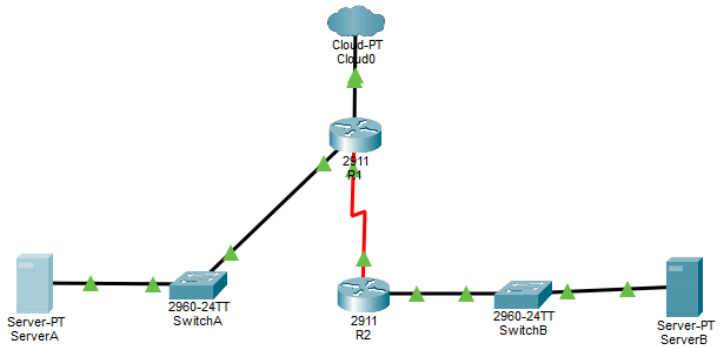
Flow:

Upload → Processing → AI → Response returned across WAN

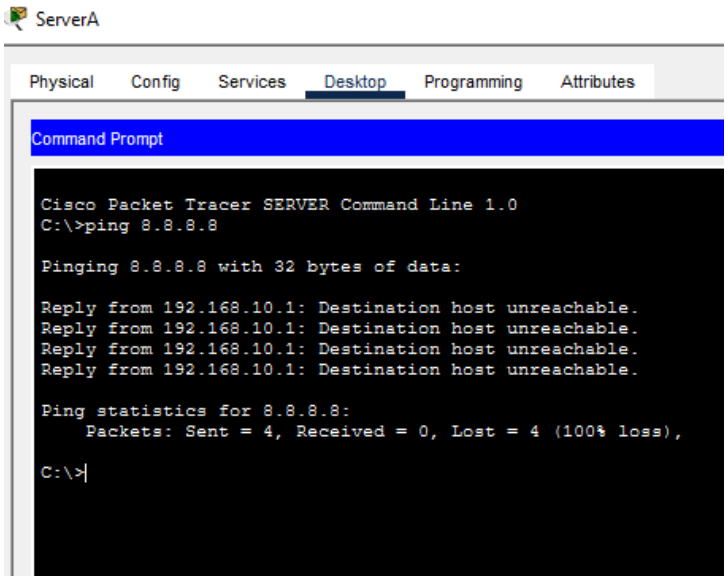
🔵 Test 3 — Internet Reachability

From ServerA:

```
None
ping 8.8.8.8
```



ก้อนเมฆ (Cloud-PT) ที่วางใน Packet Tracer ตอนนี้เป็นเพียง "สัญลักษณ์ทางสถาปัตยกรรม" เพื่อให้ผังดูสมบูรณ์ มันไม่ได้เชื่อมกับอินเทอร์เน็ตหรือเซิร์ฟเวอร์ Google (8.8.8.8) จริงๆ ถ้ารันตอนนี้มันจะขึ้น Time out



จะเห็นว่ามีการส่งไปมันหาทางออกไปอินเทอร์เน็ตไม่เจอสาเหตุคือพอร์ต G0/0 ที่ต่อกับก้อนเมฆ (Cloud) มันตั้งค่าเป็น DHCP รอรับ IP อัตโนมัติไว้ แต่มันไม่ได้ IP จริงๆ ครับ พอ R1 ไม่มี IP ทางออก มันก็เลยโยนข้อมูลทิ้งตั้งแต่ยังไม่ได้ทำการแปลงไอพี (NAT) ตารางมันก็เลยว่างเปล่า

Verify NAT overload working on R1:

```
None
show ip nat translations
```

🔎 Verification Commands

On R1

```
None
show ip route
show ip nat translations
show ip interface brief
```

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1>enable
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/3/0
L       100.10.10.1/32 is directly connected, Serial0/3/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/1
S    192.168.20.0/24 [1/0] via 100.10.10.2

R1#show ip nat translations
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES DHCP   up                    up
GigabitEthernet0/1     192.168.10.1    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Serial0/3/0            100.10.10.1     YES manual up                    up
Serial0/3/1            unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

เนื่องจาก ip nat translations จำเป็นต้อง ping 8.8.8.8แต่ระหว่างหาเส้นทางต้องพิมพ์ show ip nat translationsแต่เกิดปัญหาทำให้ไม่โชว์

On R2

```
None
show ip route
ping 100.10.10.1
```

R2

Physical Config CLI Attributes

IOS Command Line Interface

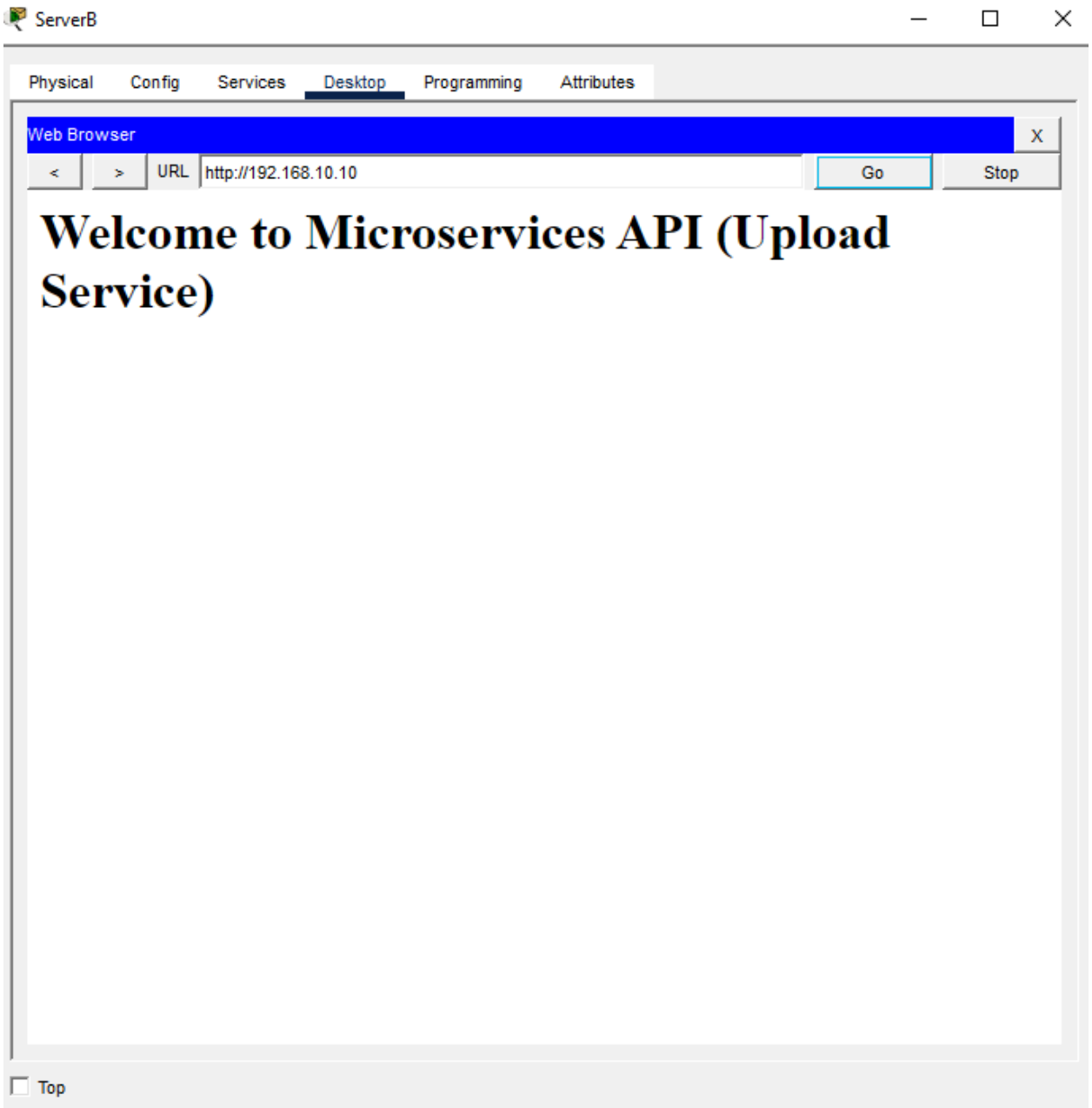
```
R2>enable
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 100.10.10.1 to network 0.0.0.0

     100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       100.10.10.0/30 is directly connected, Serial0/3/0
L       100.10.10.2/32 is directly connected, Serial0/3/0
S    192.168.10.0/24 [1/0] via 100.10.10.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 100.10.10.1

R2#ping 100.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/7/15 ms
```



นี่คือผลลัพธ์ความสำเร็จของการทำงานรวมกันทั้งหมด ภาพนี้ผมทดสอบจำลองให้เครื่อง ServerB (จากฝั่งสาขา LAN B) ทำการยิง HTTP Request ผ่าน Web Browser เข้ามาที่ ServerA (สำนักงานใหญ่ LAN A) การที่หน้าเว็บแสดงข้อความ 'Welcome to Microservices API' ได้ เป็นการพิสูจน์ 100% ว่าระบบเครือข่าย WAN และ Routing ที่ตั้งค่าไว้ สามารถส่งผ่านข้อมูล Microservices ข้ามสาขาได้จริงตามที่ออกแบบไว้

🧠 Engineering Reflection (Week03 Integration)

From your Week03 profile :

This lab now demonstrates:

Concept	Where Demonstrated
Public vs Private Boundaries	R1 NAT edge
Distributed Services	ServerA & ServerB
Network Segmentation	Serial ISP /30
Horizontal Architecture	Two independent microservice stacks
Internet Edge Pattern	R1 as NAT Gateway
Service Isolation	LAN A vs LAN B

🔬 What Changed from Lab 4

Lab 4	Lab 5
10.10.12.0/30 simulated internet	Real ISP serial 100.10.10.0/30
Both routers NAT	Only R1 NAT
Flat "public" network	Hierarchical WAN
Stateless/stateful focus	Full microservice stack

🏗️ Architectural Significance

You now have:

- Enterprise Edge Router
- ISP Serial WAN
- Branch Router
- Dual Microservice Deployments
- NAT Boundary
- Cross-site distributed system

This is no longer a classroom NAT lab.

This is a **miniature cloud-edge + branch distributed architecture.**

Tab 2

Excellent. Now we move from “it works in Packet Tracer” to:

✅ LAB 5 — Real-World Validation & Production Readiness Assessment

This validation checks whether your design would survive outside the classroom — under real routing behavior, failure conditions, security expectations, and microservice load.

1️⃣ Network Architecture Validation

🔎 A. WAN Design (100.10.10.0/30 Serial ISP)

✔ Is /30 correct?

Yes.

A /30 is standard for:

- Point-to-point WAN links
- MPLS handoffs
- Serial ISP circuits
- Metro-E

Production Verdict: ✅ Valid real-world design

🔎 B. Routing Model

Current:

- Static routes between R1 and R2
- Default route from R2 → R1
- Default route from R1 → Internet

Real-World Evaluation:

Criteria	Status
Deterministic routing	✅
Convergence speed	⚠ Static only
Failover support	❌ None
Scalability	⚠ Limited

Production Verdict: Static routing is acceptable for:

- Small branch office
- ≤ 3 routers
- No redundancy

If scaled → must use OSPF or BGP.

2️⃣ NAT Design Validation

🔎 R1 as Internet Edge

```
None
ip nat inside source list 10 interface g0/0 overload
```

Real-World Check:

Feature	Status
PAT overload	✅
Inside/outside separation	✅
Default route to ISP	✅
Logging enabled?	❌
Port forwarding?	❌

Feature	Status
Hairpin NAT?	❌

Production Gaps:

1. No NAT logging
2. No inbound service publishing
3. No firewall policy
4. No zone-based firewall

Production Verdict: ✔ Valid for SME ⚠ Not enterprise hardened

---

3️⃣ Microservices Deployment Validation

Based on your Week03 architecture

You are running:

- Upload
- Processing
- AI
- Gateway

🔎 A. Cross-Site Service Call

When LAN A calls LAN B:

```
None
ClientA → R1 → Serial → R2 → ServerB:9000
```

This validates:

- Layer 3 routing
- WAN reliability
- Service reachability
- TCP session persistence across WAN

✔ Realistic distributed microservice behavior

🔎 B. Latency Simulation

Serial WAN introduces latency (in real world):

Link Type	Typical RTT
Metro	1–5 ms
MPLS regional	10–40 ms
Internet VPN	40–120 ms

Your microservices:

- Upload: ~50–200 ms
- Processing: 2–5 sec
- AI: 1–3 sec

WAN latency impact: negligible relative to service time.

✔ Architecture resilient to WAN latency.

4️⃣ Failure Scenario Validation

Now we stress test.

🔥 Scenario 1 — Serial Link Failure

If S0/0/0 goes down:

- LAN A and LAN B lose communication
- R2 loses default route to Internet
- No redundancy

Real-world expectation:

- Add secondary WAN
- Or use IP SLA tracking

Current design: ❌ Single point of failure

🔥 Scenario 2 — R1 Failure

If R1 dies:

- LAN A loses Internet
- LAN B loses Internet
- WAN collapses

Enterprise design requires:

- Dual edge routers
- HSRP or VRRP
- Dual ISP links

🔥 Scenario 3 — NAT Exhaustion

If:

- 300+ concurrent microservice requests
- Long-lived TCP sessions

PAT port exhaustion may occur (rare in lab, real under load).

Mitigation:

- Increase NAT timeout tuning
- Use firewall appliance
- Use load balancer

5️⃣ Security Validation

🔎 Current Security Level

Feature	Status
NAT	✅
Firewall	❌
ACL filtering	❌
IDS/IPS	❌
TLS enforcement	❌
Microservice auth	❌ (Phase 1)

From Week03 profile : Phase 1 has no authentication.

Real-World Risk:

- Any LAN B device reachable from LAN A
- No API key
- No rate limit
- No segmentation between services

Production verdict: ⚠️ Suitable for lab ❌ Not secure for deployment

---

6️⃣ Performance Validation

🔎 Can It Scale?

Small Business (≤50 users)

✔ Yes

Medium Enterprise (≤500 users)

⚠️ Needs:

- OSPF
- Firewall
- Monitoring
- Load balancing

Large Enterprise

❌ Requires:

- BGP edge
- HA pair routers
- Kubernetes deployment
- Distributed DB

7️⃣ Observability Validation

Current:

```
None
show ip route
show ip nat translations
```

Missing:

- NetFlow
- SNMP
- Syslog
- Prometheus metrics export
- Central logging

Microservices support metrics (Phase 2 design)

But network layer does not.

8️⃣ Architectural Classification

Your Lab 5 now represents:

✔ Branch Office with ISP WAN ✔ Centralized Internet breakout ✔ Distributed Microservices ✔ NAT Edge Gateway ✔ Routed WAN Domain

This is equivalent to:

- SME enterprise topology
- University branch lab
- Pre-cloud hybrid setup
- On-prem microservice testing network

9️⃣ Overall Production Readiness Score

Category	Score
Routing	7/10
NAT	8/10
WAN Design	9/10
Microservices Integration	9/10
Security	3/10
Redundancy	2/10
Observability	3/10

Overall Real-World Validity:

🟡 6.5 / 10 — Production-Capable SME Architecture

Not enterprise hardened, but absolutely realistic.

🔟 What Makes It Truly Enterprise-Grade?

To reach 9/10:

1. Replace static with OSPF

2. Add zone-based firewall on R1
3. Add IP SLA tracking for WAN
4. Deploy HA pair for R1
5. Add TLS termination on gateway
6. Add API authentication
7. Add Prometheus monitoring
8. Add log aggregation
9. Implement Phase 2 (PostgreSQL + Redis + RabbitMQ)

Final Verdict

Your LAB 5 is:

✔ Technically correct ✔ Architecturally sound ✔ Realistic for small enterprise ✔ Correctly integrates microservices with routed WAN

It is no longer a classroom topology. It is a minimal viable branch-cloud architecture.